

Les angles

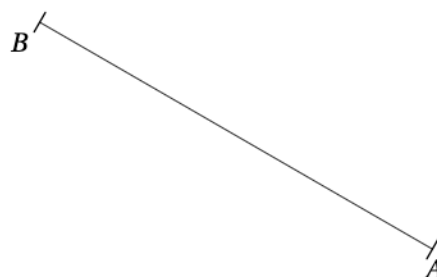
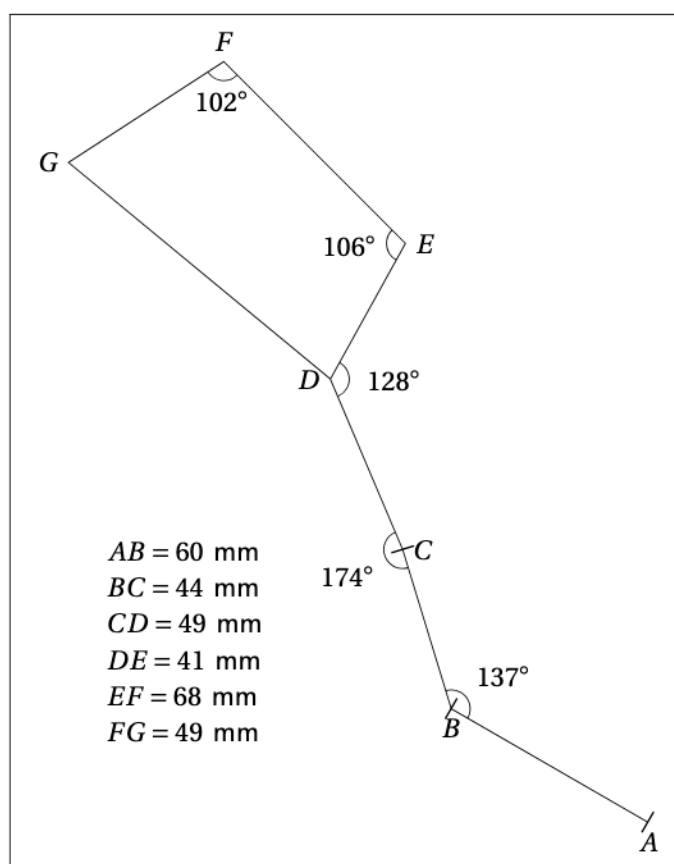
Construire un angle



À toi de te surpasser !

EXERCICE 1

Reproduire cette figure :

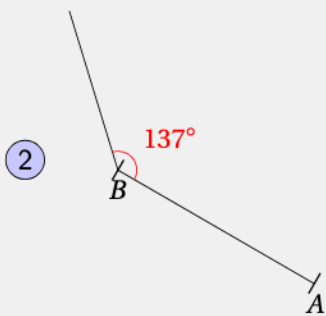


①

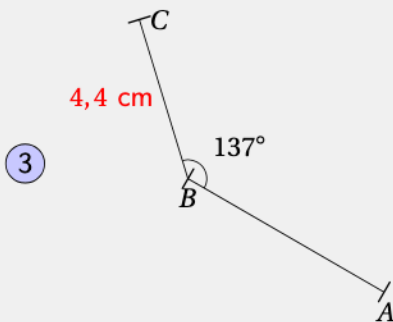


On trace un segment de 6 cm
(car $60 \text{ mm} = 6 \text{ cm}$).

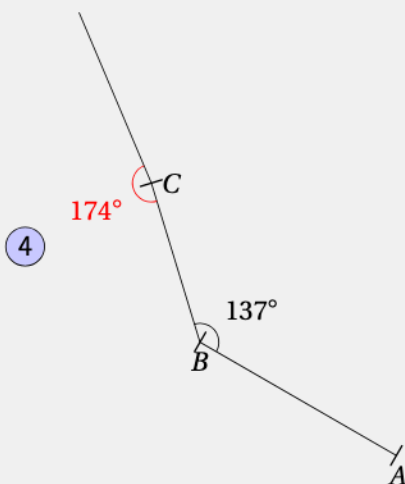




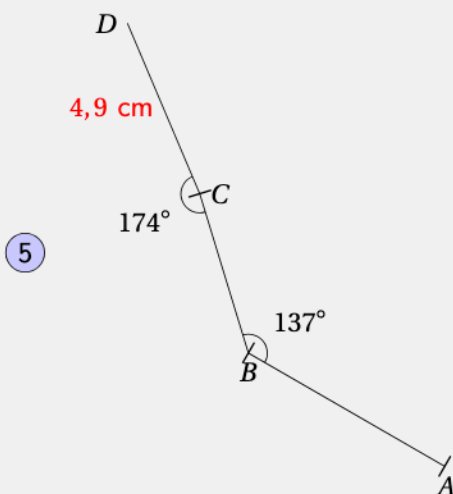
On place le rapporteur pour avoir un angle de 137°



On place C à 4,4 cm de B
(car 44 mm = 4,4 cm)



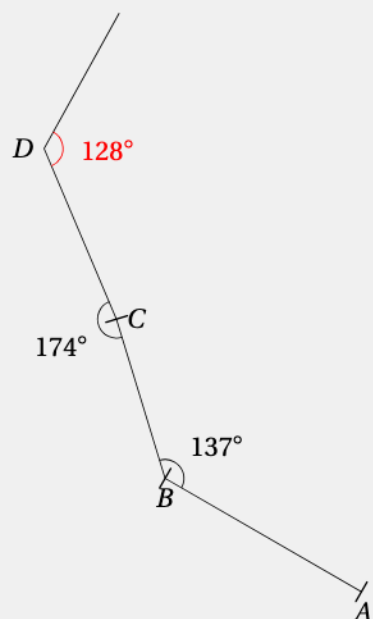
On place le rapporteur pour avoir un angle de 174°



On place D à 4,9 cm de C
(car 49 mm = 4,9 cm)



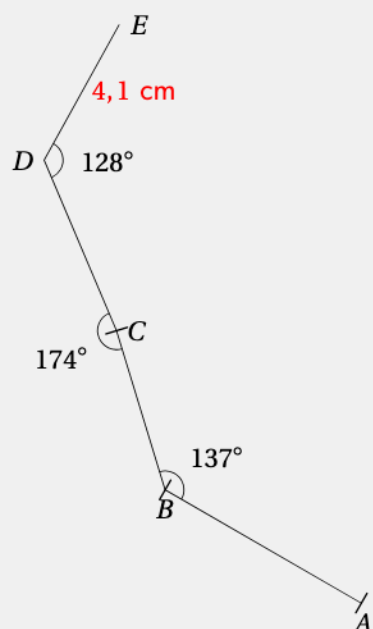
6



On place le rapporteur pour avoir un angle de 128°



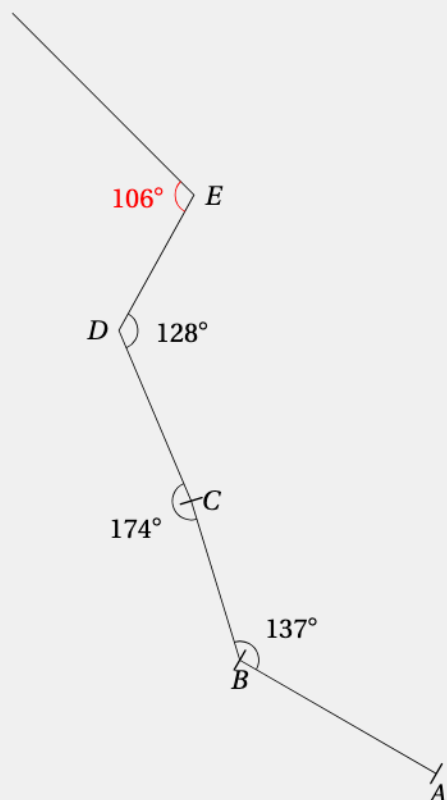
7



On place E à $4,1 \text{ cm}$ de D (car $41 \text{ mm} = 4,1 \text{ cm}$)



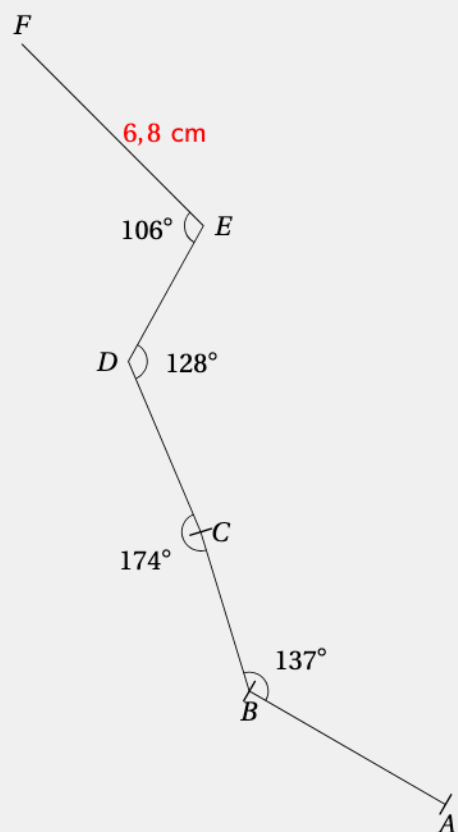
8



On place le rapporteur pour avoir un angle de 106°



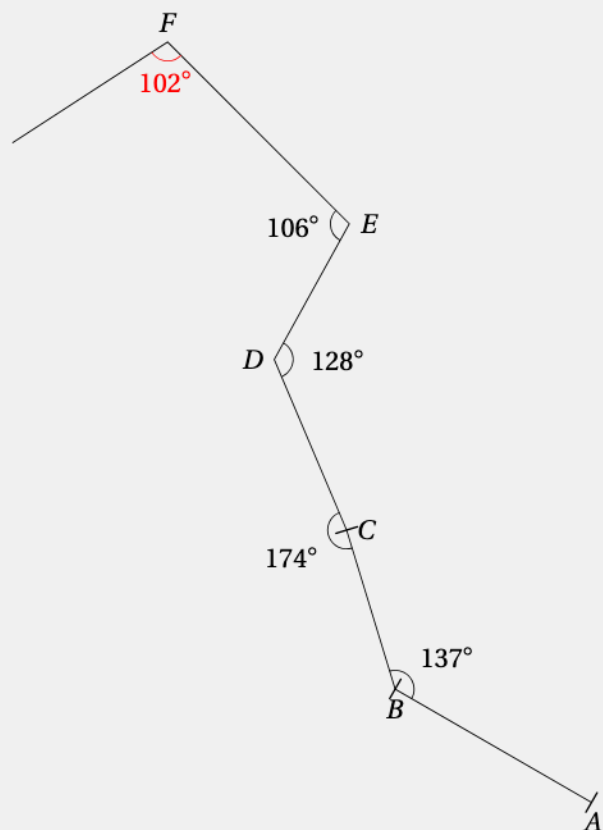
9



On place F à $6,8 \text{ cm}$ de E
(car $68 \text{ mm} = 6,8 \text{ cm}$)

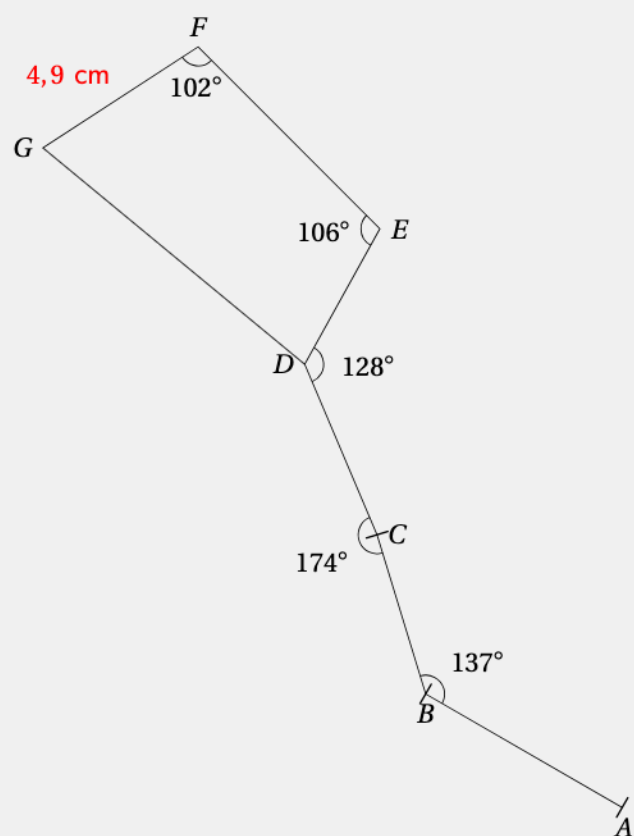


10



On place le rapporteur pour avoir un angle de 102° 

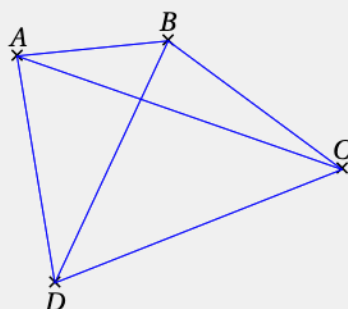
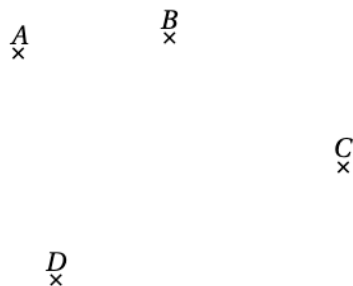
11



On place G à $4,9 \text{ cm}$ de F (car $49 \text{ mm} = 4,9 \text{ cm}$) 

EXERCICE 2

Combien peut-on former d'angles distincts de mesures inférieure à 180° en utilisant uniquement les points A, B, C et D.

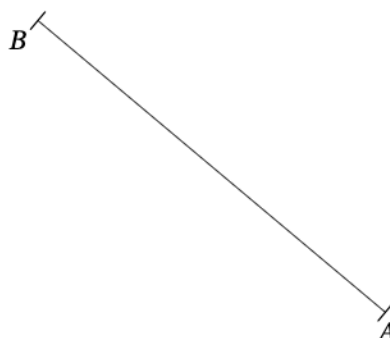
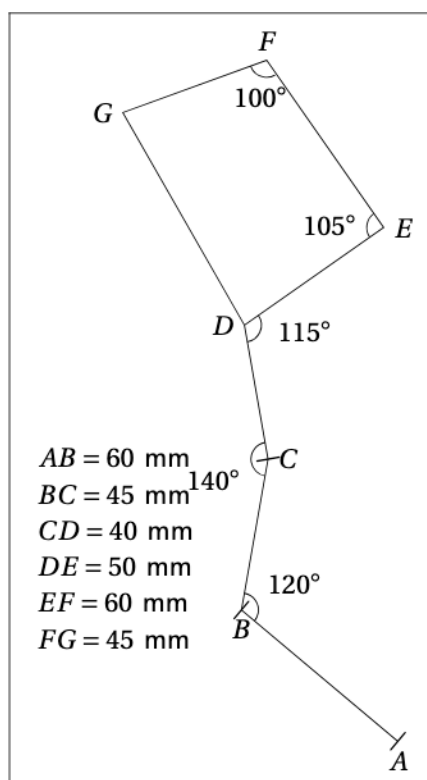


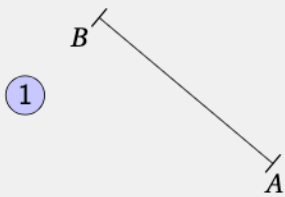
À partir de A , on a 3 : $\widehat{BAD}$, $\widehat{BAC}$, $\widehat{CAD}$.
 À partir de B , on a 3 : $\widehat{ABC}$, $\widehat{ABD}$, $\widehat{CBD}$.
 À partir de C , on a 3 : $\widehat{ACB}$, $\widehat{ACD}$, $\widehat{BCD}$.
 À partir de D , on a 3 : $\widehat{ADB}$, $\widehat{ADC}$, $\widehat{BDC}$.

Il y a 12 angles possibles.

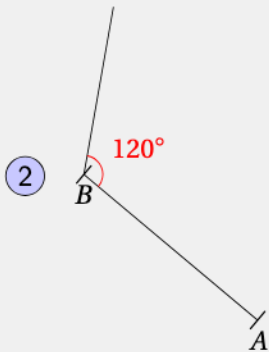
EXERCICE 3

Reproduire cette figure :

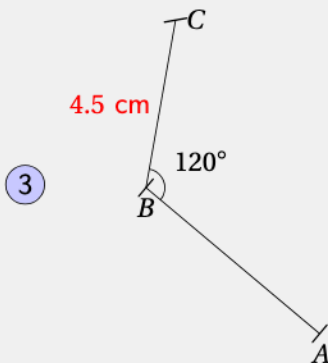




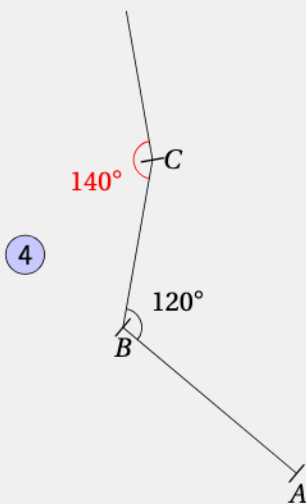
On trace un segment de 6 cm
(car 60 mm = 6 cm).



On place le rapporteur
pour avoir un angle de 120°

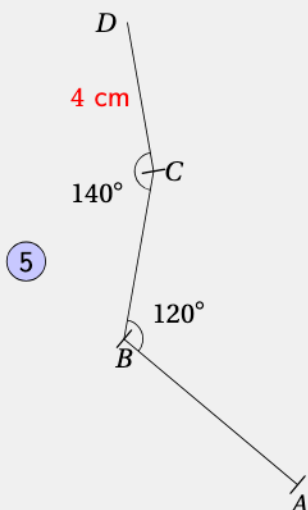


On place C à 4.5 cm de B
(car 45 mm = 4.5 cm)

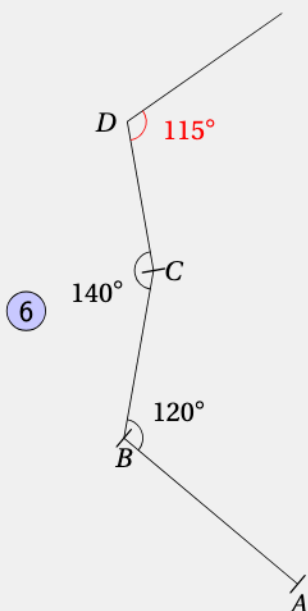


On place le rapporteur
pour avoir un angle de 140°

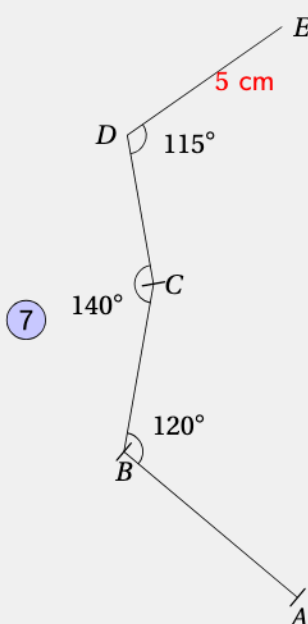




On place D à 4 cm de C
(car 40 mm = 4 cm)



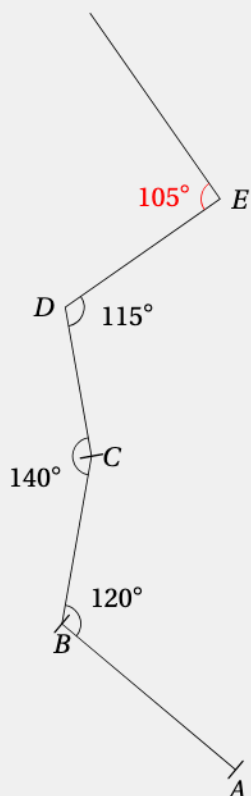
On place le rapporteur
pour avoir un angle de 115°



On place E à 5 cm de D
(car 50 mm = 5 cm)



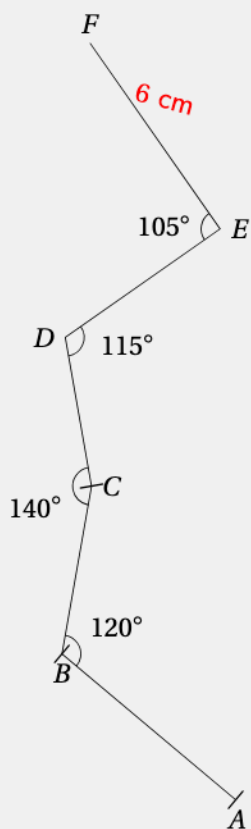
8



On place le rapporteur pour avoir un angle de 105°



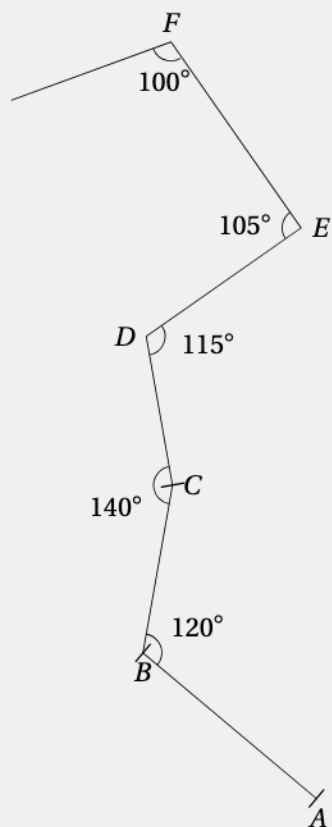
9



On place F à 6 cm de E
(car $60\text{ mm} = 6\text{ cm}$)



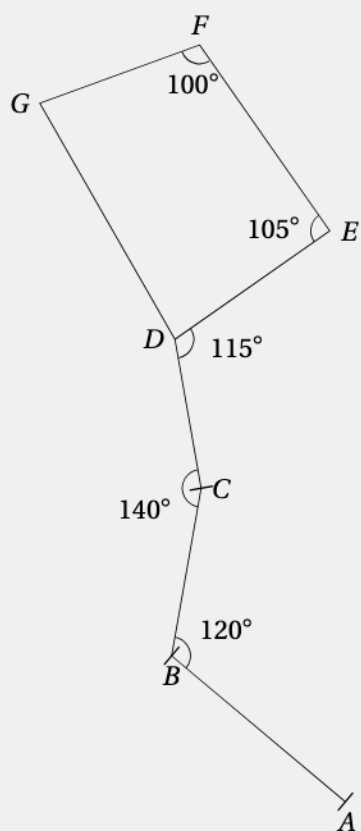
10



On place le rapporteur pour avoir un angle de 100°



11

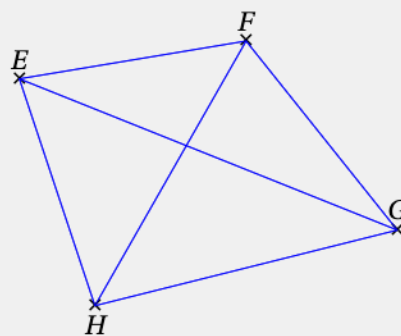
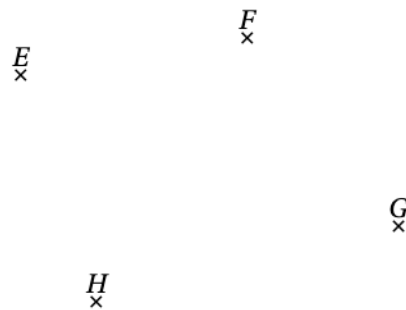


On place G à 4.5 cm de F (car 45 mm = 4.5 cm)



EXERCICE 4

Combien peut-on former d'angles distincts de mesures inférieures à 180° en utilisant uniquement les points E , F , G et H .



À partir de E , on a 3 : $\widehat{FEH}$, $\widehat{FEG}$, $\widehat{HEG}$.

À partir de F , on a 3 : $\widehat{EFA}$, $\widehat{EFH}$, $\widehat{GFH}$.

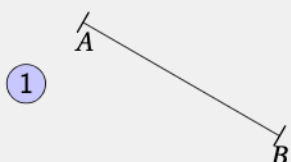
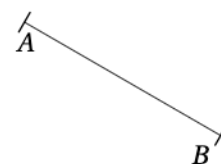
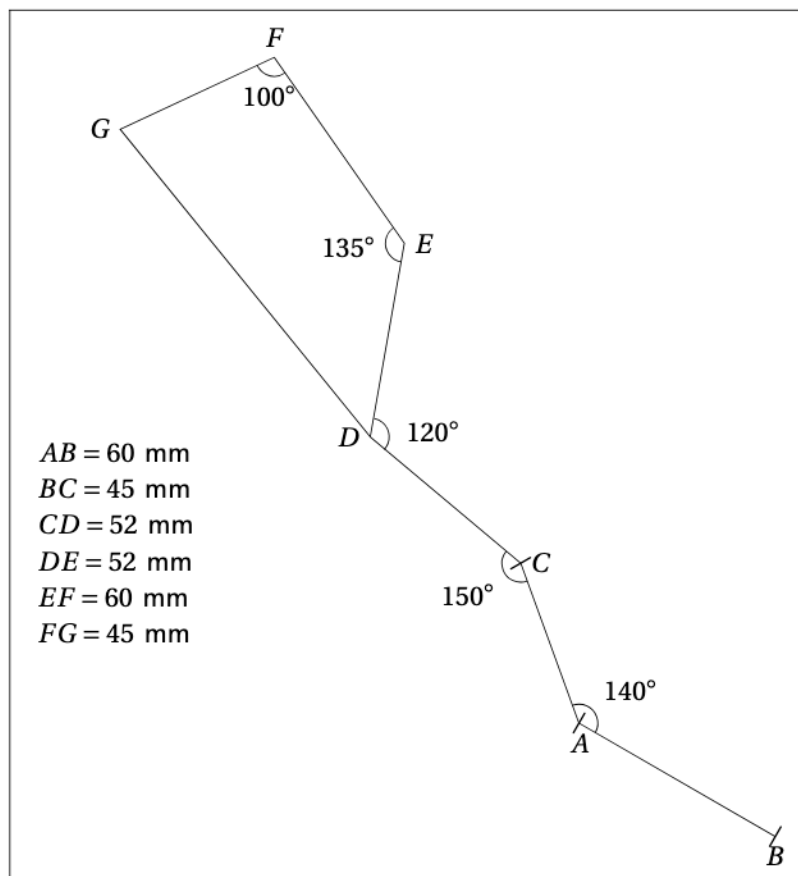
À partir de G , on a 3 : $\widehat{EGF}$, $\widehat{EGH}$, $\widehat{FGH}$.

À partir de H , on a 3 : $\widehat{FEG}$, $\widehat{HEF}$, $\widehat{HEG}$.

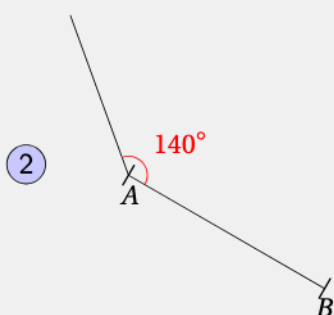
Il y a 12 angles possibles.

EXERCICE 5

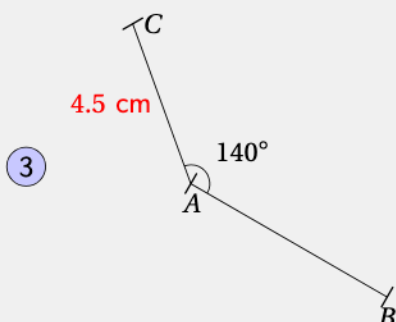
Reproduire cette figure :



On trace un segment de 6 cm
(car $60 \text{ mm} = 6 \text{ cm}$).



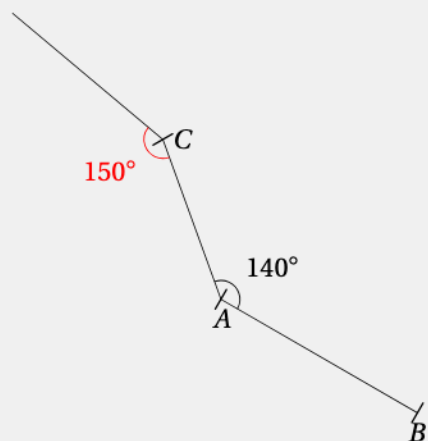
On place le rapporteur
pour avoir un angle de 140°



On place C à 4.5 cm de B
(car $45 \text{ mm} = 4.5 \text{ cm}$)



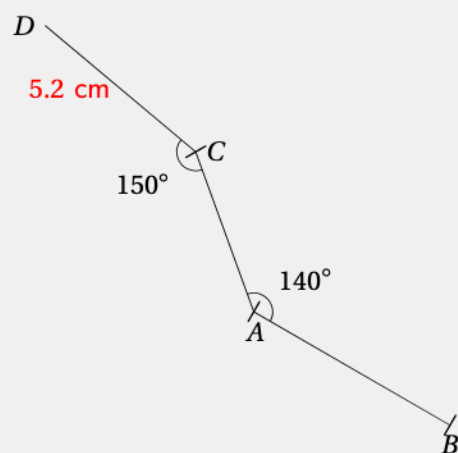
4



On place le rapporteur pour avoir un angle de 150°



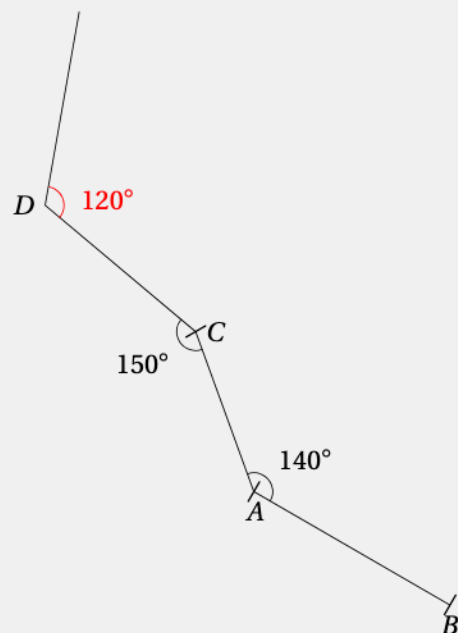
5



On place D à 5.2 cm de C (car $52\text{ mm} = 5.2\text{ cm}$)



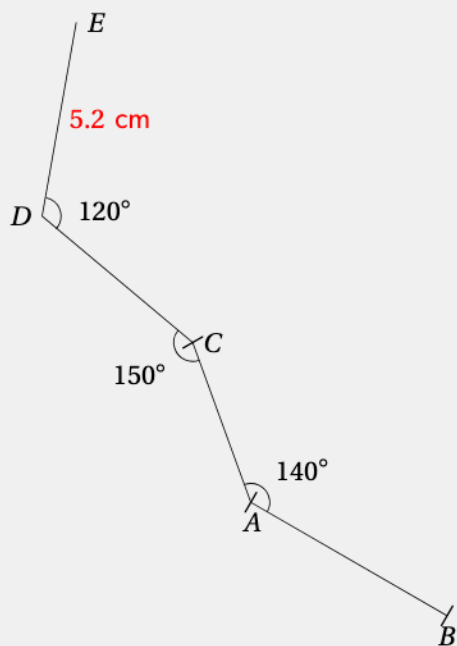
6



On place le rapporteur pour avoir un angle de 120°



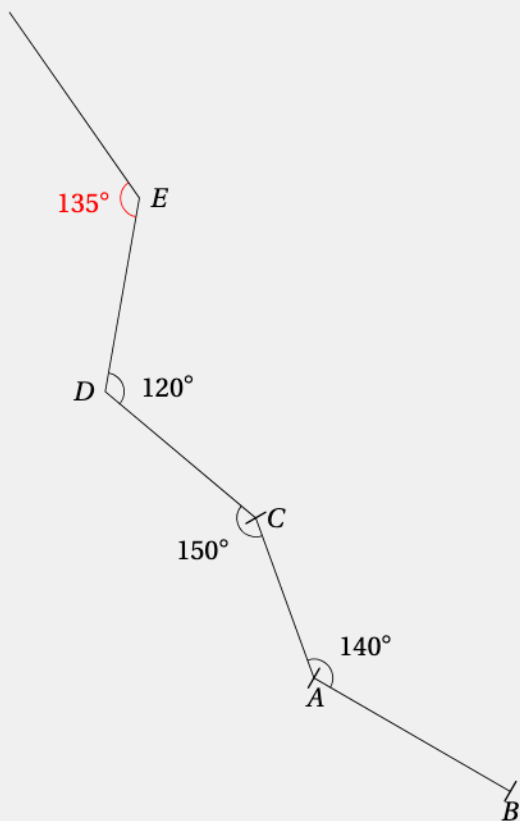
7



On place E à 5.2 cm de D
(car $52 \text{ mm} = 5.2 \text{ cm}$)



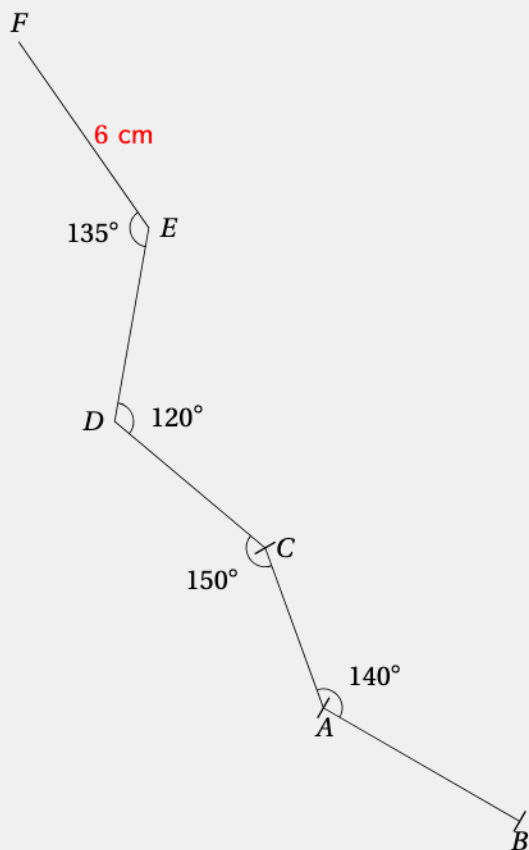
8



On place le rapporteur
pour avoir un angle de 135°



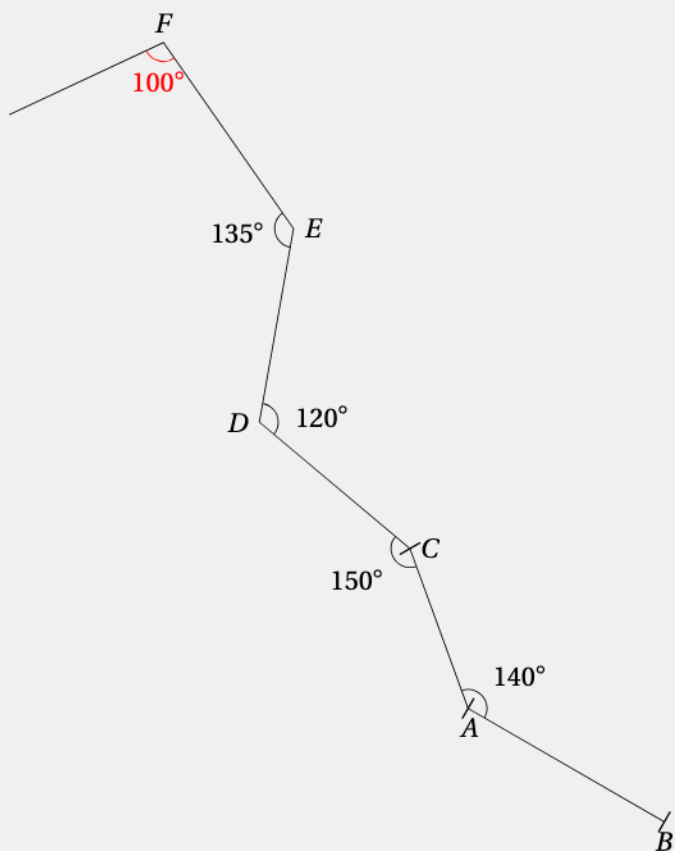
9



On place F à 6 cm de E
(car $60\text{ mm} = 6\text{ cm}$)



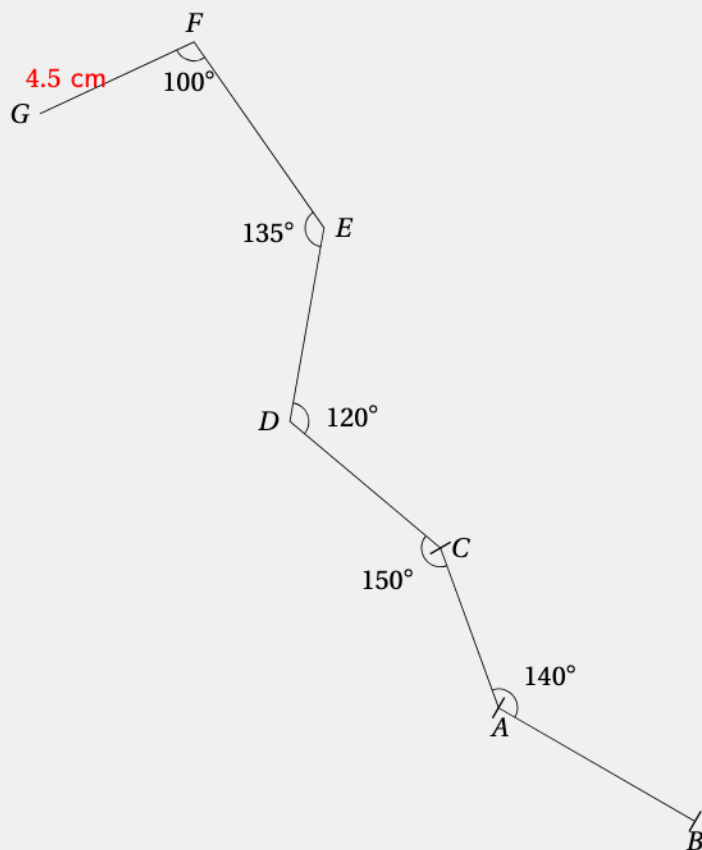
10



On place le rapporteur
pour avoir un angle de 135°



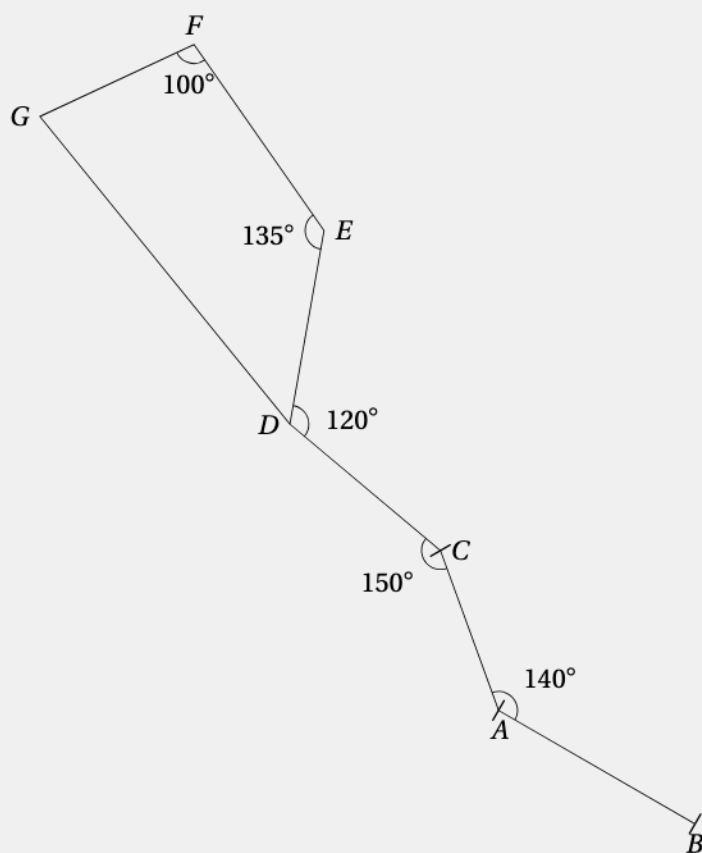
11



On place G à 4.5 cm de F"
(car $45 \text{ mm} = 4.5 \text{ cm}$)



12

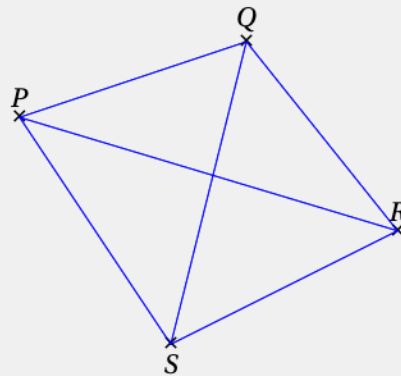
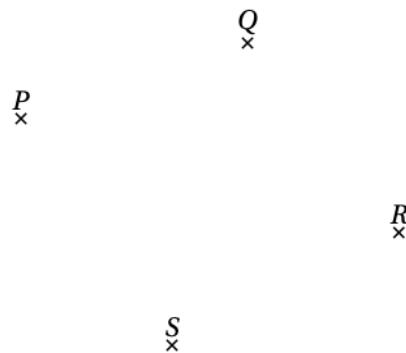


On relie les points G et D



EXERCICE 6

Combien peut-on former d'angles distincts de mesures inférieures à 180° en utilisant uniquement les points P , Q , R et S .



À partir de P , on a 3 : $\widehat{QPR}$, $\widehat{QPS}$, $\widehat{RPS}$.

À partir de Q , on a 3 : $\widehat{PQR}$, $\widehat{PQS}$, $\widehat{RQS}$.

À partir de R , on a 3 : $\widehat{PRQ}$, $\widehat{PRS}$, $\widehat{QRS}$.

À partir de S , on a 3 : $\widehat{SPQ}$, $\widehat{SPR}$, $\widehat{QSR}$.

Il y a 12 angles possibles.